

DATABASE MANAGEMENT SYSTEMS (DBMS)

Course Code: 23CS201 • Semester: 2-1 • Department of Computer Science & Engineering

Course Scope & Objectives: Relational data model, SQL queries, normalization, ACID transaction processing, concurrency control, and indexing strategies.

Unit 1: Introduction to Database Systems & ER Modeling

Key Syllabus Topics: Database Architecture, 3-Schema Architecture, Data Independence, Entity-Relationship (ER) Model, Entity Types, Relationship Constraints (Cardinality, Participation), Enhanced ER (EER) Specialization/Generalization.

Theoretical Framework & Concepts:

DBMS abstracts physical storage from application logic through physical and logical data independence. ER modeling constructs conceptual schematics using entities, attributes (composite, multi-valued, derived), and relationship cardinalities.

Implementation / Key Formula Reference:

```
CREATE TABLE Department (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(100) NOT NULL  
);
```

Frequently Asked University Exam Questions:

• **Q1: Explain 3-Schema Architecture and Data Independence.**

Solution: A: Physical (internal), Conceptual (logical), and External (view) levels. Logical independence protects views from schema changes; Physical independence shields schema from storage hardware changes.

Unit 2: Relational Model, Relational Algebra & SQL

Key Syllabus Topics: Relational Model Concepts, Integrity Constraints (Primary Key, Foreign Key, Domain, Referential), Relational Algebra Operators (Select, Project, Join, Union, Difference), Complex SQL Queries, Joins, Aggregation, Nested Subqueries, Views.

Theoretical Framework & Concepts:

Relational algebra provides procedural foundation for relational query execution. Referential integrity guarantees that foreign key values match valid primary keys in parent tables.

Implementation / Key Formula Reference:

```
SELECT d.dept_name, COUNT(s.id) as student_count
FROM Department d
JOIN Student s ON d.dept_id = s.dept_id
GROUP BY d.dept_name
HAVING COUNT(s.id) > 50;
```

Frequently Asked University Exam Questions:

• **Q1: Explain different types of SQL Joins with examples.**

Solution: A: INNER JOIN returns matching rows; LEFT JOIN returns all left rows plus matched right rows; RIGHT JOIN returns all right rows; FULL OUTER JOIN returns all rows from both tables.

Unit 3: Functional Dependencies & Relational Normalization

Key Syllabus Topics: Anomalies in Unnormalized Tables (Insertion, Deletion, Update), Functional Dependencies (Armstrong Axioms), 1NF, 2NF (Full FD), 3NF (Transitive FD Elimination), BCNF (Strict Determinant Rule), Multi-Valued Dependencies & 4NF.

Theoretical Framework & Concepts:

Normalization reduces data redundancy and eliminates update anomalies without information loss. A table is in 3NF if for every $X \rightarrow Y$, X is a superkey or Y is prime. BCNF strictly requires X to be a superkey for every non-trivial FD.

Implementation / Key Formula Reference:

```
-- BCNF condition: For all  $X \rightarrow Y$ ,  $X$  must be a candidate key
```

Frequently Asked University Exam Questions:

• **Q1: Differentiate between 3NF and BCNF with a suitable scenario.**

Solution: A: 3NF permits non-superkey determinants if the dependent attribute is prime. BCNF strictly forbids non-superkey determinants, eliminating all functional redundancy.

Unit 4: Transaction Processing and Concurrency Control

Key Syllabus Topics: ACID Properties (Atomicity, Consistency, Isolation, Durability), Transaction States, Serializability (Conflict & View), Recoverability, Lock-Based Protocols (2-Phase Locking - 2PL, Strict 2PL), Deadlock Handling (Prevention, Detection, Wait-Die, Wound-Wait), Timestamp-Based Ordering.

Theoretical Framework & Concepts:

Transactions are atomic units of execution. Concurrency control guarantees isolation. 2-Phase Locking ensures conflict serializability by dividing locks into Growing (acquiring locks) and Shrinking (releasing locks) phases.

Implementation / Key Formula Reference:

```
BEGIN TRANSACTION;
UPDATE Accounts SET balance = balance - 500 WHERE id = 101;
UPDATE Accounts SET balance = balance + 500 WHERE id = 102;
COMMIT;
```

Frequently Asked University Exam Questions:

- **Q1: Explain ACID properties in DBMS with real-world banking examples.**

Solution: A: Atomicity (all-or-nothing transfer), Consistency (total money remains constant), Isolation (intermediate states hidden from concurrent users), Durability (committed data persists despite server crash).

Unit 5: Storage, Indexing, and Query Optimization

Key Syllabus Topics: File Organization (Heap, Sequential, Hash), Primary Index, Secondary Index, Clustering Index, B-Tree and B+ Tree Indexing Structure, Query Processing Steps, Cost Estimation, Heuristic Query Optimization.

Theoretical Framework & Concepts:

B+ Trees provide balanced multi-level indexing where all data records/pointers reside at leaf nodes linked sequentially, facilitating both random point lookups and high-speed range scans in $O(\log N)$ disk I/O operations.

Implementation / Key Formula Reference:

```
CREATE INDEX idx_student_dept ON Student(dept_id, semester);
```

Frequently Asked University Exam Questions:

- **Q1: Why are B+ Trees preferred over B-Trees for database indexing?**

Solution: A: Leaf nodes in B+ Trees are linked sequentially for fast range scans; internal nodes store only routing keys, yielding higher fan-out and shallower tree depths.